Math Kangaroo Practice

Grade 3-4 · 20 questions · harder · seed 44932

Part A: Each question is worth 3 points

1. A bar code is made of 23 black and white bars, black and white taking turns. The first and the last bar are black. Each black bar is either wide or narrow. There are 9 more white bars than wide black bars.

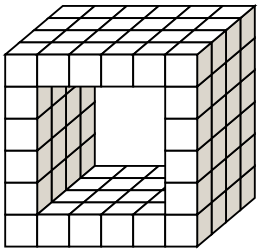
How many narrow black bars are there?

A. 2 ***B.*** 3 ***C.*** 8 ***D.*** 9 ***E.*** 10

2. The numbers 34 and 142 have the same sum of digits (3 + 4 = 7 and 1 + 4 + 2 = 7). What is the first number greater than 7092 whose sum of digits is the same as for 7092?

A. 7128 ***B.*** 7119 ***C.*** 7102 ***D.*** 7101 ***E.*** 7093

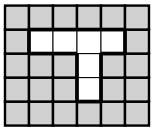
3. Leo built this tunnel out of identical cubes. The hole goes straight through from the front to the back.

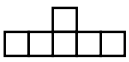
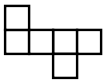
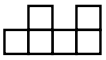
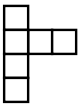


How many cubes did Leo use?

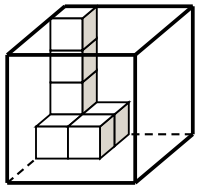
A. 144 ***B.*** 100 ***C.*** 96 ***D.*** 80 ***E.*** 20

4. Which piece covers the empty area exactly? The pieces may be turned and flipped over.



A.  ***B.***  ***C.***  ***D.*** ***E.*** 

5. Mia puts cubes into a glass box, as in the picture. Every cube stands on the bottom or on another cube.



How many cubes are in the box?

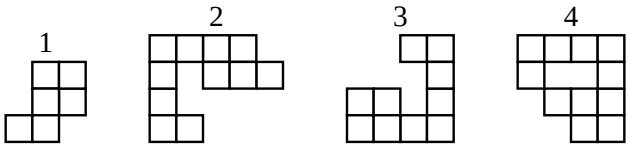
A. 9 ***B.*** 8 ***C.*** 7 ***D.*** 6 ***E.*** 4

6. Omar, who is older than Leo by 3 years and 2 days, was born on August 31, 2001.
When was Leo born?

A. August 29, 2004 ***B.*** September 2, 1998 ***C.*** September 2, 2004
D. August 31, 2004 ***E.*** September 2, 2005

Part B: Each question is worth 4 points

7. Which two of these pieces fit together to make a rectangle? Pieces may be turned and flipped over.



A. 1 and 3 ***B.*** 1 and 2 ***C.*** 2 and 3 ***D.*** 3 and 4 ***E.*** 1 and 4

8. Ben has 4 different T-shirts, 4 different pairs of trousers and 2 different pairs of shoes. An outfit is one T-shirt and one pair of trousers and one pair of shoes.
How many different outfits can Ben make?

A. 16 ***B.*** 32 ***C.*** 33 ***D.*** 37 ***E.*** 64

9. Sara thinks of a number, divides by 3, then takes away 7, then multiplies by 3, and gets 9.
Which number did Sara think of?

A. 9 ***B.*** 27 ***C.*** 30 ***D.*** 32 ***E.*** 35

10. Mia reads a chapter from page 93 to page 118 (both included).
How many digits are printed in the page numbers of this chapter?

A. 71 ***B.*** 73 ***C.*** 79 ***D.*** 87 ***E.*** 246

11. Leo has the cards 1, 2 and 7. Leo must use every card exactly once, with +, –, ×, ÷ and brackets.
Which number can Leo *not* get?

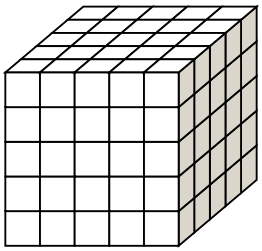
A. 7 ***B.*** 12 ***C.*** 15 ***D.*** 16 ***E.*** 17

12. Mia writes the word BANANA. Then Mia crosses out every letter that also appears in the word PLANET.

How many different letters are not crossed out?

A. 4 ***B.*** 3 ***C.*** 2 ***D.*** 1 ***E.*** 0

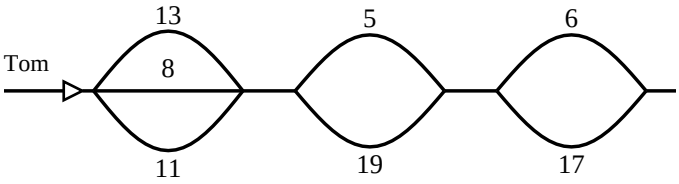
13. Tom glued 125 small cubes into a big 5 × 5 × 5 cube (like the picture) and painted the whole outside of it, also the bottom. Then the big cube falls apart.



How many small cubes have no paint at all?

A. 54 ***B.*** 36 ***C.*** 31 ***D.*** 28 ***E.*** 27

14. Tom walks along the road from left to right and adds up the numbers along the way.

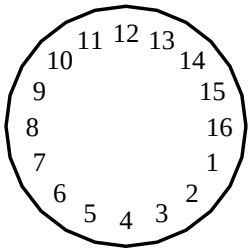


Which total can Tom *not* get?

A. 19 ***B.*** 23 ***C.*** 35 ***D.*** 38 ***E.*** 47

Part C: Each question is worth 5 points

15. A round cookie with the numbers 1 to 16 around its edge is broken by one straight crack into two pieces. The sums of the numbers on the two pieces are equal.



Where does the crack go?

A. between 13 and 14, and between 3 and 4

B. between 11 and 12, and between 3 and 4

C. between 4 and 5, and between 10 and 11

D. between 12 and 13, and between 4 and 5

E. between 2 and 3, and between 10 and 11

16. Manon, Emma, Chloe, Nicolas and Hana each play a different instrument: the drum, the trumpet, the piano, the harp and the guitar. Nicolas doesn't play the guitar. Hana does not blow into the instrument. Emma plays with sticks. Nicolas plays an instrument with strings. Manon presses keys.
Which statement is **not** true?

A. Nicolas plays the guitar ***B.*** Hana plays the guitar ***C.*** Manon plays the piano
D. Chloe plays the trumpet ***E.*** Emma plays the drum

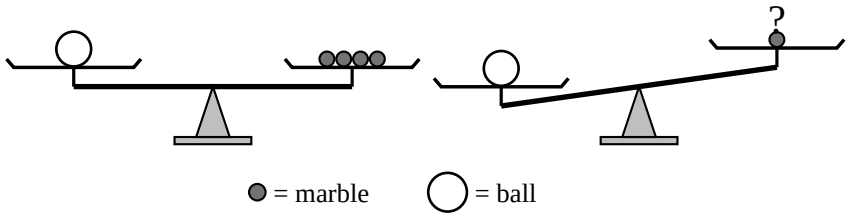
17. Cleo's new rug is 2 m 7 cm long. Four of these lengths are the same as that.
Which one is *not* the same?

A. 2 m 70 mm ***B.*** 207 cm ***C.*** 2070 mm ***D.*** 20 dm 7 cm ***E.*** 2 m 7 mm

18. The odometer of a scooter shows 1269 km. Each digit is bigger than the digit before it.
After how many more kilometres will this happen again?

A. 20 ***B.*** 18 ***C.*** 9 ***D.*** 8 ***E.*** 1

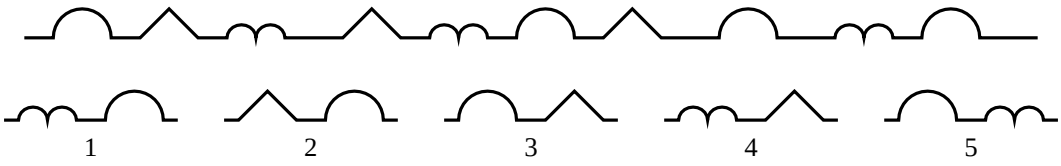
19. The first balance is level. On the second balance there is 1 ball on the left and 1 marble on the right.



How many marbles must be added to the right plate to make it level?

A. 7 ***B.*** 6 ***C.*** 5 ***D.*** 4 ***E.*** 3

20. Far away we see the skyline of a castle.



How many of the pieces 1–5 below belong to the line?

A. 0 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 5

Answer Key

1.	**E**	**2.**	**B**	**3.**	**D**	**4.**	**E**	**5.**	**C**	**6.**	**C**
7.	**A**	**8.**	**B**	**9.**	**C**	**10.**	**A**	**11.**	**E**	**12.**	**D**
13.	**E**	**14.**	**B**	**15.**	**D**	**16.**	**A**	**17.**	**E**	**18.**	**C**
19.	**E**	**20.**	**B**								

Solutions

1. E (10) — Black is at both ends, so there are 12 black and 11 white bars. Wide: 11 – 9 = 2. Narrow: 12 – 2 = 10. [2003 Q17 · narrow_bars]

2. B (7119) — 7 + 0 + 9 + 2 = 18. Going up from 7092, the first number with digit sum 18 is 7119 (7 + 1 + 1 + 9 = 18). [2007 Q4 · next_same_sum]

3. D (80) — The front has 6 × 6 – 4 × 4 = 20 cubes. The tunnel is 4 cubes long, so 20 × 4 = 80. [2002 Q20 · count_tunnel]

4. E — The empty area has 6 squares; all pieces do too, so look at the shape. Turn piece E and it fits exactly; each of the others has one square in a different place. [2003 Q18 · one_piece]

5. C (7) — Count each column from the bottom: 1 + 4 + 1 + 1 = 7. [2007 Q14 · count_inside]

6. C (September 2, 2004) — Older means born earlier, so Leo was born 3 years and 2 days *after* August 31, 2001: 3 years later is August 31, 2004, then 2 more days: September 2, 2004. [2007 Q12 · younger_birthday]

7. A (1 and 3) — Pieces 1 and 3 have 16 squares together and fit into a 4 × 4 rectangle. No other pair does. [2007 Q8 · two_make_rectangle]

8. B (32) — Each T-shirt goes with each pair of trousers and each pair of shoes: 4 × 4 × 2 = 32. [2002 Q22 · outfits]

9. C (30) — Go backwards: 9 ÷ 3 = 3, 3 + 7 = 10, 10 × 3 = 30. [2007 Q5 · think_of_a_number]

10. A (71) — There are 26 pages: 7 × 2 + 19 × 3 = 71 digits. [2003 Q16 · from_page]

11. E (17) — (2 – 1) × 7 = 7; (7 – 1) × 2 = 12; 1 + 2 × 7 = 15; (1 + 7) × 2 = 16. But with 1, 2 and 7 nothing gives 17 (the closest are 16 = (1 + 7) × 2 and 21 = (1 + 2) × 7). [2002 Q21 · which_cannot]

12. D (1) — Crossed out: A and N (they are in PLANET). Left: B, so 1 different letters. [2007 Q3 · only_in_first]

13. E (27) — They are the cubes inside: 3 × 3 × 3 = 27. [2003 Q9 · painted]

14. B (23) — 19 = 8 + 5 + 6; 35 = 13 + 5 + 17; 38 = 13 + 19 + 6; 47 = 11 + 19 + 17. No route gives 23. [2007 Q1 · impossible_total]

15. D (between 12 and 13, and between 4 and 5) — Total 136, so each piece has 68: 13+14+15+16+1+2+3+4 = 68 and 5+6+7+8+9+10+11+12 = 68. [2002 Q11 · straight_crack]

16. A (Nicolas plays the guitar) — The only possible matching: Manon – piano; Emma – drum; Chloe – trumpet; Nicolas – harp; Hana – guitar. [2002 Q18 · which_not_true]

17. E (2 m 7 mm) — 2 m 7 cm = 207 cm = 2070 mm. But 2 m 7 mm = 2007 mm. [2003 Q10 · not_equal]

18. C (9) — Keep as many digits on the left as possible. The next such number is 1278: 1278 – 1269 = 9. [2002 Q24 · going_up]

19. E (3) — One ball = 4 marbles, so the left plate is worth 1 × 4 = 4 marbles. The right plate has 1, so add 4 – 1 = 3. [2002 Q9 · what_to_add]

20. B (2) — Pieces 1, 3 fit. Not: 2 (it has the right shapes but in the wrong order); 4 (it has the right shapes but in the wrong order); 5 (its flat part between the two features has the wrong length). [2002 Q7 · how_many_fit]